

Andrei-Cătălin Popa

PHD STUDENT IN ASTROPHYSICS

Lennard-Jones School of Chemical and Physical Sciences, Keele University, Staffordshire, ST5 5BG, UK

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Education

Université Côte d'Azur

Nice, France

MSC IN ASTROPHYSICS AND SPACE SCIENCE (ERASMUS MUNDUS JOINT MASTER'S PROGRAMME)

Mar. 2025 – Sept. 2026

Thesis: *Characterising M-dwarf Stars as Exoplanet Hosts in Preparation for the PLATO Mission.*

Advisors: Dr. Orlagh Creevey (OCA), Dr. Valentina D'Orazi (Tor Vergata), Dr. Ulrike Heiter (Uppsala).

Research Internship: *Fundamental Parameters of FGK Stars.*

Advisors: Dr. Orlagh Creevey (OCA) and Dr. Caroline Soubiran (Bordeaux).

Università degli Studi di Roma "Tor Vergata"

Rome, Italy

MSC IN ASTROPHYSICS AND SPACE SCIENCE (ERASMUS MUNDUS JOINT MASTER'S PROGRAMME) — FIRST SEMESTER

Sept. 2024 – Mar. 2025

Consortium of four universities: Università degli Studi di Roma "Tor Vergata", Universität Bremen, Université Côte d'Azur, and University of Belgrade.

University College London

London, United Kingdom

MSCI IN PHYSICS (FIRST CLASS HONOURS)

Sept. 2018 – Jun. 2022

Supervisor: Prof. Daisuke Kawata (Mullard Space Science Laboratory).

Thesis: *Hunting for Intermediate-Mass Black Hole Binaries.*

Research Experience

Astrophysics Research Centre

Keele, United Kingdom

PHD RESEARCH: CALIBRATING STELLAR MODELS WITH CHEMICAL TRACERS OF INTERNAL MIXING PROCESSES

Oct. 2026 – present

- Supervised by Dr. Pierre Maxted. Research within the ESA PLATO mission consortium, focused on using chemical abundance tracers in binary stars to calibrate stellar structure and evolution models.
- Investigating internal mixing processes (convective overshooting, atomic diffusion, rotation) through precision constraints provided by binary star systems and PLATO lightcurves.

Observatoire de la Côte d'Azur

Nice, France

MASTER'S THESIS: CHARACTERISING M-DWARF STARS AS EXOPLANET HOSTS

Jan. 2026 – Sept. 2026

- Supervised by Dr. Orlagh Creevey (OCA), Dr. Ulrike Heiter (Uppsala University) and Dr. Valentina D'Orazi (Tor Vergata), within the Erasmus Mundus Joint Master's in Astrophysics and Space Science (MASS).
- Deriving luminosities and radii of benchmark M-dwarf stars via spectral energy distribution (SED) fitting and bolometric flux calculations, using Gaia XP spectra and multi-band photometry.
- Building and testing Python pipelines for photometric and spectroscopic data comparison, integrating literature stellar parameters for cross-validation.
- Contributing to PLATO host-star modelling by refining fundamental parameters used in the mission's stellar characterisation framework.

Astrophysics Research Institute

Liverpool, United Kingdom

SUMMER RESEARCH PROJECT: WHITE DWARF COOLING SEQUENCES AND INFRARED COLOUR EXCESS

Jul. 2025 – Sept. 2025

- Supervised by Prof. Maurizio Salaris. Investigated whether the apparent broadening of the white dwarf cooling sequence in NGC 6397 reflects an intrinsic IR-excess population or statistical scatter.
- Analysed JWST and HST photometric data; constructed synthetic white dwarf populations from theoretical isochrones and a Salpeter IMF, with Monte Carlo uncertainty propagation.
- Developed Python routines to compare observed and simulated colour-magnitude diagrams, applying Gaussian Mixture Models (GMM) and Bayesian Information Criterion (BIC) analysis.

Scholarships & Honours

2024–2026 **Erasmus Mundus (EMJM) Scholarship**, European Education and Culture Executive Agency

€33,600

2022 **Summer Research Internship Scholarship**, Dept. of Physics & Astronomy, UCL

£2,000

2022 **"Teach Physics" Internship Award**, Ogden Trust — Eastbury Community School, London

£1,400

Teaching

Eastbury Community School

Barking, United Kingdom

SUMMER PHYSICS TEACHING INTERN — OGDEN TRUST "TEACH PHYSICS" PROGRAMME

Jun. – Jul. 2022

Taught A-level and GCSE physics through interactive sessions and lab exercises; created classroom materials aligned with UK physics curricula; supported Science Club activities.

Outreach

UCL Observatory

DEMONSTRATOR

Mill Hill, United Kingdom

Oct. 2019 – Sept. 2022

Trained to operate robotic and manual telescopes (14-inch Schmidt–Cassegrain; twin 24/18-inch refractors). Led public observing sessions for school groups and members of the public; developed science communication skills across diverse audiences.

Department of Physics & Astronomy, University College London

London, United Kingdom

OUTREACH INTERN

Jun. – Sept. 2022

Organised outreach activities in local schools and community groups; planned and delivered science communication events including UCL's public festival *Your Universe*; produced materials translating astrophysical research for non-specialist audiences.

Service & Committees

2024–2025 **Student Representative**, Quality Evaluation Committee, Erasmus Mundus MASS Programme

Rome/Nice

Skills & Professional Memberships

Programming

Python, MATLAB, IDL

Astronomical Software

TOPCAT, SAO DS9, AstroImageJ, VOSA, iSpec

Tools & Environments

Git, $\LaTeX$, UNIX/Linux

Professional Memberships

Institute of Physics (IOP); Mensa Romania; England Handball Referee Association

Languages

English: Fluent | **Romanian:** Native | **French:** Intermediate